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Paleomagnetic results from the Upper Silurian of the Shan-Thai-Malay block, southwest Yunnan, China

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Abstract. 77 paleomagnetic samples from the Upper Silurian reddish limestone and marlstone of the Shan-Thai-Malay block near Baoshan (99.1°E, 24.8°N), western Yunnan, China, reveal a pre-folding remnant magnetization, with a mean direction of D=49.6°, I= -3.2°, k=9.7, a_{95}=5.5°, corresponding a paleopole at 207.0°E, 36.8°N. These results, unlike the upper Paleozoic data with steeper inclinations published earlier, suggest that the block was located at equatorial latitude during late Silurian time, probably adjacent to the northern margin of Gondwanaland.

Introduction

The Three-River region is a tectonic collage of displaced fragments from Gondwana and Cathaysia continents during Paleotethys evolution stage(Mitchell, 1981; Stauffer, 1983; Zhong, 1995). The Shan-Thai-Malay block, or Baoshan block in Chinese part, has long been referred to being a piece of continental fragment from Gondwana, indicated by the Carboniferous tillite and Gondwana fauna (Ren et al., 1980; Cao, 1986). While the Simao block and Linchang Arc in the eastern vicinity, demarcated by the Changning-Menglian melange Zone (Main Paleotethys Suture Zone), are interpreted to be rifted blocks from the Cathaysia continent (Hsu, 1994) in the Late Paleozoic. Nevertheless, these blocks are still poorly constrained by paleomagnetic data, especially the Lower Paleozoic. The data published for the Shan-Thai-Malay block are mainly derived from the Upper Devonian, Carboniferous and Lower Permian (McElhinny et al., 1974; Fang et al., 1989; Huang and Opdyke, 1991) and these results locate the block at southern hemisphere, probably adjacent to Australia during the Late Paleozoic. Several recent papers postulated that the majority of the Chinese continental blocks were derived from Paleozoic Gondwanaland (e.g., Burrett et al., 1990). The early Paleozoic paleomagnetic results from Australia suggest that Australia was located at northern intermediate to low latitude from the Cambrian to Silurian and afterwards moving southward (Goleby, 1981; Piper, 1987). Till Devonian time, it shifted to southern low latitude. The lower Paleozoic paleomagnetic data from the Yangtze and North China blocks also show an equatorial inclination (Lin et al., 1985a; Zhao et al., 1993). Similarities in Cambrian trilobites, early Paleozoic fossil assemblages, phosphorites as well as Precambrian stratigraphy between South China (including southwest) and Australia reinforce the explanation by paleomagnetists (Lin et al., 1985b). To test this hypothesis, it is critical to determine paleolatitude of the block and paleogeographic relationships with other Chinese blocks by means of paleomagnetism.

Geology, sampling and measurements

The Paleozoic and Triassic sequence in the Baoshan Block is a continuous marine succession with a total thickness over 8000m (Geological Bureau of Yunnan, 1990). The folding took place in late Triassic to Jurassic time when marine sedimentation was ceased and continental reddish clastic rocks started accumulation. An unconformity is widely seen between the Jurassic and underlain marine strata. The folding was most probably caused by the collision event of the Baoshan Block with the Lincang arc terrene in the east and Central Burma Block in the west. The folded Paleozoic sedimentary cover of the block, as a foreland fold and thrust zone of the neighboring orogenic belts, is well exposed in the region. The Upper Silurian is composed of marine carbonates, and dominated by purple limestone and marlstone in the upper part with abundant bioclasts and burrows as well as small ripple laminations, which indicate a shallow water and oxidized environment. The age of the strata is well determined by conodont fossils of *Spathognothodus sp., S. steinhornensis? eosteinhornensis, Trichchonodella syminertrica,*

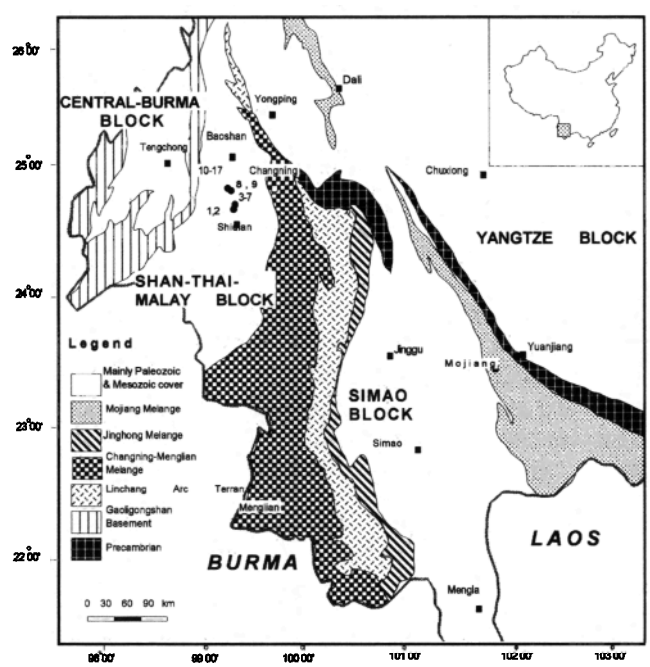


Figure 1. Simplified tectonic map of the Three-River region showing geographic locations of the major tectonic units and localities of paleomagnetic sampling. Numerals by solid circles are site numbers.

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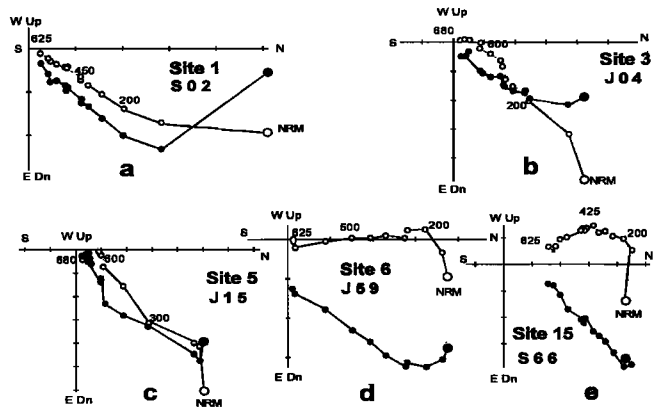


Figure 2. Orthogonal projections of magnetization vector end points during thermal demagnetization in geographic coordinates for representative samples from southern sites (a, b, c) and northern sites(d, e) respectively. Numbers beside symbols represent temperature (°C). Closed (open) symbols refer to the projection on horizontal (vertical) planes.

and Cystoidea of *Camarocrinus asiaticus* (Geological Bureau of Yunnan, 1990). A total of 103 oriented core samples from 17 sites are taken with a portable drill from these carbonates at two quarries and two natural outcrops from two localities near Shidian County (99.1°E, 24.8°N), about 50 km south of Baoshan city (Fig. 1). The opposite dipping directions of the beds (55° / 42° for 6 sites at southern location, and 250° / 45° for the northern sites) in the two locations about 20 km away from each other make a fold test possible. Lithologies sampled are slightly different: sites 1 to 7 come from purple marlstone, and sites 8 and 9 were taken from light gray limestone, and sites 10 to 17 are from reddish limestone.

Samples from nine sites were processed at the paleomagnetic laboratory of ETH at Zurich with a 2G cryogenic magnetometer, while another eight sites were measured by a 2G 755-R cryogenic magnetometer at the paleomagnetic laboratory of Chinese Academy of Sciences at Beijing. The results produced by the two laboratories are completely consistent. Stepwise thermal demagnetization method was used for all samples. The principal component analysis (PCA) technique of Kirschvink (1980) was employed for analyzing the demagnetization data. For those samples without linearly decaying to the origin but poling to remagnetization circles during thermal treatment, great circle method was used to evaluate the magnetic directions (McFadden and McElhinny, 1988).

Results

Intensities of natural remnant magnetization (NRM) of most samples range from 3.5×10^{-4} A/m to 1.1×10^{-3} A/m. Progressive thermal demagnetization revealed that almost all the samples, except for those from sites 8 and 9 which are weak and decay rapidly in intensities and become unstable above 350°C, show double components magnetization (Fig. 2). The low temperature component (LTC), which generally accounts for less than 20% of the initial NRM intensity, and can be defined below 200 °C, is a recent viscous overprint acquired during transportation or storage with random magnetization directions. The high temperature component (HTC) in about one third of the samples appears above 300 °C and reaches the origin of vector plots at around 650 °C. While another two third of the samples either do not decay linearly to the origin or become unstable above 600 °C. Their vector changes, however, follow a great circle though with relatively small best-fit arc length, giving a possibility for evaluating better constrained directions. The site-mean directions were determined utilizing set points by PCA and sector-constrained great circle intersection (McFadden & McElhinny, 1988) (Fig. 3). Only a few samples from the northern sites show a little more complicated magnetization history when looking at inclination changes (Fig. 2e). The intermediate temperature component shown in Fig. 2e is similar in deviated declination to the Paleogene direction (Chen et al., 1995) and was probably formed by the Tertiary remagnetization. All samples from 15 sites are unblocked at temperatures from 625°C to 680°C, indicating that hematite is regarded as the prime remanence carrier, as estimated by reddish color of the rocks.

Characteristic remnant magnetization (ChRM) directions are either northeasterly downward (sites from the south) or upward (from the northern sites), both with shallower inclinations. They are not completely coincident after tilt-correction probably because the overprinted component could not be fully cleaned for the southern sites or smaller best-fit arc length causing intersections away from the mean. ChRM directions of the HT component are mainly northeasterly upward with shallow inclinations. Application of the fold test on the site-mean directions of the ChRM component from all samples (sites 8 and 9 from light gray limestone as well as a few probably misoriented samples with ChRM far from the mean direction are excluded for further calculation) yields a positive result (Fig. 4): a sharp increase of precision parameter (k) from 2.6 to 9.7 by giving unit weight to N=77 samples ($k_2/k_1=3.7$) or from 2.6 to 18.6 by unit weight to N=7 sites ($k_2/k_1=7.15$) corresponds to a positive fold test at more than 99% confidence (McElhinny, 1964). These ChRM directions are apparently prefolding in origin, and probably represent a original remanence, with mean D=49.6°, I=-3.2°, a_{95}=5.5°, k=9.7, n=77 samples, corresponding a pole position at 207.0°E, 36.8°N.

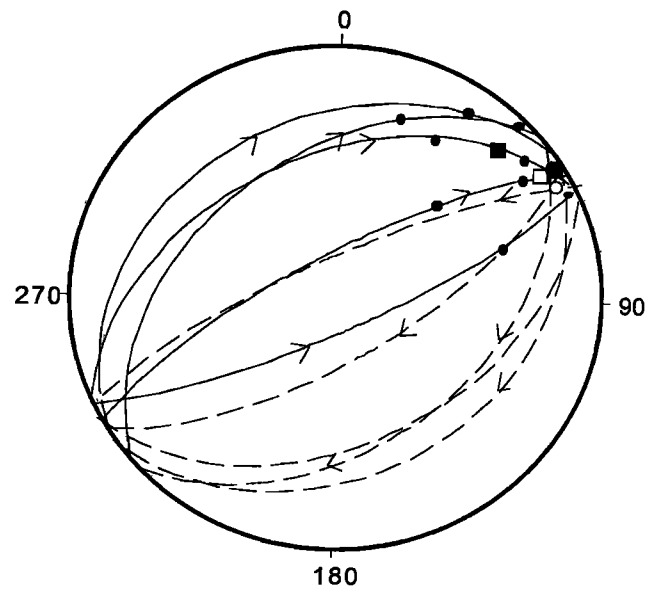


Figure 3. Equal-area projection of one set point and 5 remagnetization great circles from Site S01. Closed square is the set point, and open square is the site mean direction. Small closed and open circles refer to the lower and upper hemisphere, and the larger closed circle is the mean direction of 77 samples.

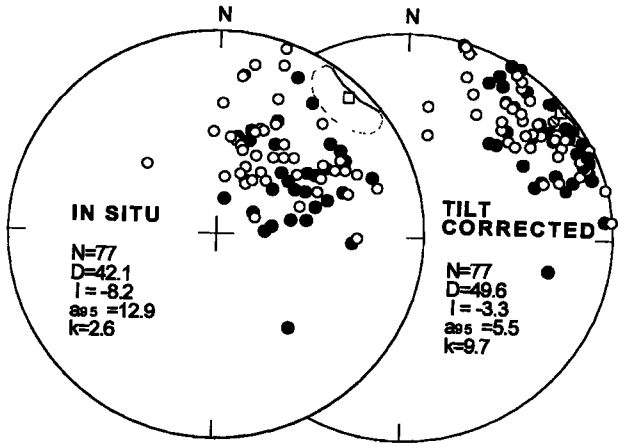


Figure 4. Equal-area projection of ChRM directions from 77 samples of the Upper Silurian of the Shan-Thai-Malay block in western Yunnan. Closed (open) symbols represent directions plotted onto the lower (upper) hemisphere.

Discussion and conclusion

Both overprinted and primary components are revealed in the paleomagnetic results from this study. The LT component is interpreted to be a secondary magnetization imposed in recent overprints. Since the folding in the study area took place mainly in the Jurassic, as indicated by the unconformity between the Middle Jurassic red beds and underlain marine Paleozoic, the prefolding magnetization isolated from the ChRM component must have been acquired at least before the Jurassic. The fold test for ChRM directions from the two locations shows a maximal concentration at about 130% correction. This deviation could probably be resulted either from slightly different sedimentation periods of the two locations which have not been better constrained by fossil records, or the diagenetic chemical remanants from the both locations were acquired at slightly different times. The pole position calculated from these data is well away from both Devonian and Carboniferous poles of the same region published earlier (Fig. 5) (Fang et al., 1989; Huang and Opdyke, 1991), reinforcing the interpretation of primary origin. Widespread remagnetizations during the Meso-Cenozoic have been reported from northern Thailand (Chen and Courtillot, 1989; Yang and Besse, 1993). Our other Paleozoic samples also show a widespread remagnetization during the block collisions in the late Mesozoic and early Cenozoic, and these results will be presented elsewhere. The Silurian samples from this study were indeed suffered a heavy overprint which was most probably imposed in the Cenozoic when India and Eurasia collided. The reason for survival of primary magnetization from this study is probably related to the marly lithology of the samples that could more favorable for preventing the thermal fluids induced by the nearby orogenesis to reach the rock, and this interpretation certainly needs to be further investigated. The Upper Silurian pole from this study is far away from the Yangtze pole, while closer to the North China pole (Fig. 5), implying large post-Silurian relative motions between these blocks.

It has been a long argument that the Shan-Thai-Malay block was rifted from the northern margin of Gondwanaland in late Late Paleozoic (Mitchell, 1981; Stauffer, 1983; Metcalfe, 1988). This postulation has been supported by a series of similarities in Paleozoic fauna, Early Permian cool-water brachiopods and foraminifers, and late Paleozoic glacio-marine diamictites of the region with Australia. The Upper Silurian results obtained from this study demonstrate a quite shallow inclination, as reported from other Chinese blocks such as North China, Yangtze (Lin et al., 1985a; Liu et al., 1984; Opdyke et al., 1987; Zhao et al., 1993). The lower Paleozoic paleomagnetic results from Australia suggest that Australia was located at northern intermediate to low latitude from the Cambrian to Silurian and continuously moving southward (Piper, 1987; Goleby, 1981). Till Devonian time, it shifted to southern low latitude. The Silurian north pole of Australia is very close to the Baoshan Block and other Chinese blocks (see Table 1), implying that they might be once closely related. The Chinese blocks, along with Australia, may have been part of northern Gondwana during early and middle Paleozoic time. By the Late Paleozoic, as the main body of Gondwanaland was shifting southward and reaching southern high latitude in Carbono-Permian time, the major Chinese blocks such as North China, Yangtze, Tarim and minor blocks in southeast China must have been rifted off and enjoyed equatorial latitudes as suggested by paleomagnetic data , fossil records as well as sedimentary environments. The Shan-Thai-Malay block had not been rifted away from Gondwanaland probably until at least Permian time, as indicated by geological, palaeontological and paleomagnetic evidences.

In summary, we interpret the characteristic remnant magnetization of the Upper Silurian samples taken from western Yunnan to be primary. The paleomagnetic inclinations determined from two sampling localities are shallow, suggesting a low paleolatitude about 1.6°. The Shan-Thai-Malay block, together with other Chinese blocks and Australia, may have been a part of northern Gondwanaland laying at equatorial latitude during late Silurian time, as suggested by both geological and paleomagnetic evidences.

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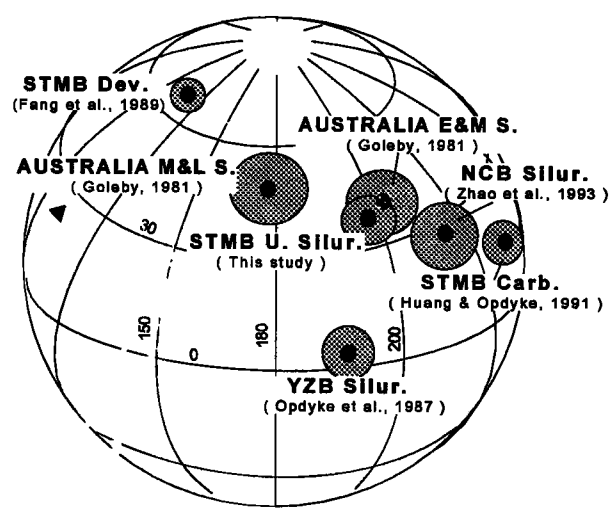


Figure 5. Paleomagnetic poles for the Upper Silurian of the Shan-Thai-Malay block (abbreviated as STMB U. Silur.) from this study, the Silurian of the North China block (NCB Silur.) from Zhao et al. (1993), the Silurian of Yangtze block (YZB Silur.) from Opdyke et al. (1986), the Devonian (STMB Dev.) from Fang et al. (1989) and Carboniferous (STMB Carb.) from Huang and Opdyke (1991) of the Shan-Thai-Malay block respectively. The Silurian Australia poles (Australia E&M S. and M&L S.) are from Goleby (1981). Closed triangle represents the sampling area.

Age	SLong	SLat	PLong	PLat	A 95	Latitude	Reference
Shan-Thai-Malay Block							
U. Silur.	99.1	24.8	207.0	36.8	5.5	3.2 S	This paper
Dev.	99.0	25.0	133.8	66.5	5.8	42	Fang et al.,1989
U. Carb.	99.3	25.2	77.7	-17.5	9.3		Huang et al.,1991
North China Block							
Silurian	105.6	37.4	228.4	34.2	8.2	6.7	Zhao et al.,1993
Yangtze Block							
Silurian	108.0	28.5	194.7	4.9	5.6	5.2	Opdyke et al.,1987
Southeast Australia							
E. Silur.	177.5	47.3	9.4	5.1			Goleby, 1981
L Silur.	214.5	38.2	8.7	4.1			Goleby, 1981

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References

Burrett, C.F., J. Long and B. Stait, Early-Middle Paleozoic biogeography of Asian terrens derived from Gondwana, in *Paleozoic Paleogeography and Biogeography*, edited by W.S. Mckerrow and C.R. Scotese, pp. 163-174, Geol. Soc. London Mem. 12, 1990.

Cao, Y.G., Paleobiogeography and tectonic evolution of western Yunnan, *Bull. Chinese. Acad. Geol. Sci.* 10, 3-25, 1986.

Chen, H.H., J.P. Dobson, F. Heller and J. Hao, Paleomagnetic evidence for clockwise rotation of the Simao block since Cretaceous: a consequence of India-Asia collision, *Earth Planet. Sci. Lett.*, 134, 203-217, 1995.

Chen, Y. and V. Courtillot, Widespread Cenozoic (?) remagnetization in Thailand and its implications for the India-Asia collision, *Earth Planet. Sci. Lett.*, 93, 113-122, 1989.

Fang, W., Van der Voo, R. and Q.Z. Liang, Devonian paleomagnetism of Yunnan Province across the Shan Thai-South China suture, *Tectonics*, 5, 939-952, 1989.

Geological Bureau of Yunnan, *Regional Geology of Yunnan Province*, Geol. Publ. House, Beijing, 727, 1990.

Goleby, B.R., Early Paleozoic paleomagnetism in Southeast Australia, in *Global Reconstruction and the Geomagnetic Field during the Paleozoic*, edited by McElhinny, M.W., DRPC, Dordrecht - Boston - London, 1981.

Hsu, K.J., *The Geology of Switzerland: An Introduction to Tectonic Facies*, pp. 192-205, Princeton Univ. Press, Princeton, New Jersey, 1994.

Huang J.Q. and B.W. Chen, *The Evolution of the Tethys in China and Adjacent Region*, pp. 78, Geol. Publ. House, Beijing, 1987.

Huang, K.N. and N.D. Opdyke, Paleomagnetic results from the Upper Carboniferous of the Shan-Thai-Malay block of western Yunnan, China, *Tectonophysics*, 192, 333-344, 1991.

Kirschvink, J.L., The least-squares line and plane and the analysis of paleomagnetic data, *Geophys. J.R. Astron. Soc.*, 62, 699-718, 1980.

Lin, J.L., M.D. Fuller and W.Y. Zhang, Preliminary Phanerozoic polar wander paths for the North and South China Blocks, *Nature*, 313, 444-449, 1985a.

Lin, J.L., M.D. Fuller and W.Y. Zhang, Paleogeography of the North and South China Blocks during the Cambrian, *J. of Geodynamics*, 2, 91-114, 1985b.

Liu, C and Q.Z. Liang, Magnetostratigraphy of the Yulongshi and Chuipingshan formations, Qujing, Yunnan, *Chinese Science Bull.*, 4, 232-234, 1986.

McElhinny, M.W., N.S. Haile and A.R. Crawford, Paleomagnetic evidence shows Malay Peninsula was not part of Gondwanaland, *Nature*, 252, 641-645, 1974.

McElhinny, M.W., Statistical significance of the fold test in paleomagnetism, *Geophys. J. R. Astron. Soc.*, 8, 338-340, 1964.

McFadden, P.L. and McElhinny, M.W., The combined analysis of remagnetization circles and direct observations in paleomagnetism, *Earth & Plant. Sci. Lett.*, 87, 161-172, 1988.

Mitchell, A.H.G., Phanerozoic plate boundaries in mainland SE Asia, the Himalayas and Tibet, *J. Geol. Soc. London*, 183, 109-122, 1981.

Opdyke, N.D., K.N. Huang, G.Z. Xu, W.Y. Zhang and D.V. Kent, Paleomagnetic results from the Silurian of the Yangtze Paraplatform, *Tectonophysics*, 139, 123-132, 1987.

Piper, J.D.A., *Paleomagentism and the Continental Crust*, Open University Press, Milton Keynes, pp. 434, 1987.

Ren, J.S., C.F. Jiang, Z.K. Zhang and D.Y. Qin, *Geotectonic Evolution of China*, pp. 203, Science Press, Beijing, 1980.

Stauffer, P.H., Unraveling the mosaic of Paleozoic crustal blocks in southeast Asia, *Geol. Rundsh.*, 72, 1061-1080, 1983.

Yang, Z.Y. and J. Besse, Paleomagnetic study of Permian and Mesozoic sedimentary rocks from Northern Thailand supports the extrusion model for Indochina, *Earth Planet. Sci. Lett.*, 117, 525-552, 1993.

Zhao, X.X., R. Coe, H.N. Wu and Z.Y. Zhao, Silurian and Devonian paleomagnetic poles from North China and implications for Gondwana, *Earth Planet. Sci. Lett.*, 117, 497-506, 1993.

Zhong, D.L., *Paleotethys--Lithosphere Tectonic Evolution of W. Yunnan and Sichuan*, Science Press, Beijing, 1995, in press.

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